XISD6329 – TASK 2: MAJOR PROJECT SUBMISSION

Dear Students,

Task 2 is the **major project submission** and includes the development, implementation, testing, and demonstration of your project solution. **Each group member must submit the same report individually on Arc.**

Azure DevOps Requirement

Each group must use **Azure DevOps throughout the development and implementation of Task 2**.

The following Azure DevOps services must be used:

- **Azure Repos** for source-code management and regular meaningful commits.
- **Azure Boards** for task allocation, progress tracking, and management of team activities.
- **Azure Pipelines** for automated builds and automated testing.
- A complete and up-to-date **README** containing the relevant project information and setup instructions.

All group members must actively participate in and contribute to the Azure DevOps project according to their allocated tasks.

Please add me to your Azure DevOps organisation/project as a **Basic user** using:

makletsoalo@rbi.ac.za

Your development must be based on the documentation submitted in **Task 1** and must incorporate the feedback provided throughout the module.

Regular physical or virtual meetings must take place between team members. Each team member is accountable for their allocated tasks. If a member neglects their responsibilities, this must be reported to the lecturer for immediate intervention and reflected in the peer evaluation.

Task 2 Requirements

You are required to:

- Develop and test the **Android mobile application**.
- Update the website where required based on the Task 1 documentation and feedback.
- Ensure that the final application **runs on a physical mobile phone and not only on an emulator**.
- Develop the code for the application's GUI.
- Develop appropriate input controls.
- Develop the required program modules.
- Allow users to **register and log in**.

- Securely encrypt or hash user passwords.
- Implement **Single Sign-On (SSO)**.
- Allow users to change appropriate settings within the application.
- Create and use the required **API**.
- Correctly integrate the API into the mobile application.
- Design and develop database tables using an appropriate DBMS.
- Populate the database using real or simulated data with **at least 10 records per applicable table**.
- Verify and test the complete system.
- Ensure that invalid user input is handled appropriately without causing the application to crash.
- Develop a help system and appropriate **user documentation**.
- Keep **Azure DevOps updated throughout the entire implementation process**.
- Conduct automated testing on the main functionality of the application.
- Configure **Azure Pipelines** to automatically build and test the application.
- Maintain regular and meaningful commits in **Azure Repos**.
- Maintain an up-to-date **README**.

Working Application

The application must be a **fully working application without errors** and must be updated based on the feedback provided throughout the course.

The application must **compile and run** for its features to receive marks.

The user interface must be appropriate, user-friendly, and consistent. The application must also handle invalid input without crashing.

Demonstration Video

You must submit a **professional demonstration video with voice-over**.

The demonstration video must:

- Demonstrate all required features of the application.
- Show the application running on a **physical mobile device**.
- Include a clear voice-over explaining the functionality being demonstrated.
- Show the data stored in the online-hosted authentication service.
- Demonstrate the API.
- Demonstrate the database and stored data.

- Demonstrate relevant Azure DevOps evidence, including project progress, commits, and automated build/testing.

Please plan ahead for the demonstration video, particularly if you do not personally own an Android device.

Do not leave physical-device testing, Azure DevOps updates, database population, automated testing, or the demonstration video until the final submission day.